is changed by changing the current, an emf is induced in that same coil (Figure 4.18). This phenomenon is known as **self-induction**. The emf induced is called self-induced emf.

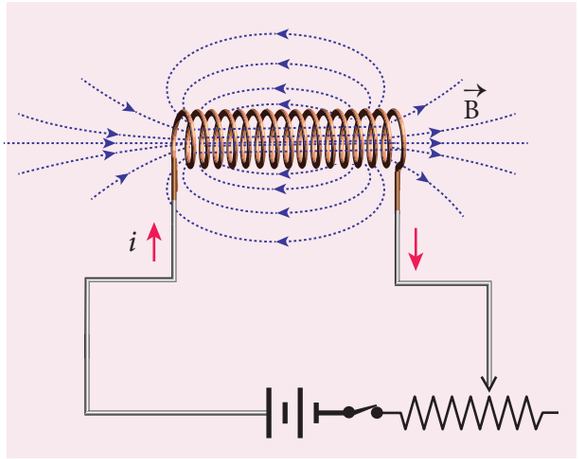



**Figure 4.18** Self-Induction

Let $\Phi_B$ be the magnetic flux linked with each turn of the coil of N turns, then the total flux linked with the coil $N\Phi_B$ (flux linkage) is proportional to the current $i$ in the coil.

$$N\Phi_B \propto i$$
$$N\Phi_B = Li \qquad (4.8)$$
$$\text{(or) } L = \frac{N\Phi_B}{i}$$

The constant of proportionality $L$ is called self-inductance or coefficient of self-induction of the coil.

If $i = 1\text{A}$, then $L = N\Phi_B$. **Self-inductance or simply inductance of a coil is defined as the flux linkage with the coil when 1A current flows through it.**

When the current $i$ changes with time, an emf is induced in it. From Faraday's law of electromagnetic induction, this self-induced emf in the coil is given by

$$\varepsilon = -\frac{d(N\Phi_B)}{dt}$$
$$= -\frac{d(Li)}{dt} \qquad \text{(using equation 4.8)}$$
$$\therefore \varepsilon = -L\frac{di}{dt} \qquad (4.9)$$
$$\text{(or) } L = \frac{-\varepsilon}{di/dt}$$

The negative sign in the above equation implies that the self-induced emf always opposes the change in current with respect to time. If $di/dt = 1\ \text{A s}^{-1}$, then $L = -\varepsilon$. **Inductance of a coil is also defined as the opposing emf induced in the coil when the rate of change of current through the coil is 1 A s$^{-1}$.**

### Unit of inductance

Inductance is a scalar and its unit is $\text{Wb A}^{-1}$ or $\text{V s A}^{-1}$. It is also measured in henry (H).

$$1\ \text{H} = 1\ \text{Wb A}^{-1} = 1\ \text{V s A}^{-1}$$

The dimensional formula of inductance is $M\,L^2\,T^{-2}A^{-2}$.

If $i$ = 1 A and $N\Phi_B = 1$ Wb turns, then $L = 1\text{H}$.

Therefore, **the inductance of the coil is said to be one henry if a current of 1 *A* produces unit flux linkage in the coil.**

If $di/dt = 1\text{A s}^{-1}$ and $\varepsilon = -1\ \text{V}$, then $L = 1\text{H}$.

Therefore, **the inductance of the coil is one henry if a current changing at the rate of 1 A s$^{-1}$ induces an opposing emf of 1 V in it.**

### Physical significance of inductance

We have learnt about inertia in XI standard. In translational motion, mass is a measure of linear inertia; in the same way, for rotational motion, moment of inertia is a measure of rotational inertia (Refer sections 3.2.1 and 5.4 of XI physics text book). Generally, inertia means opposition to change its state.



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